

HW #1: Mixture Density Networks

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1 Model and implementation

For an input $\mathbf{x}$ and target $\mathbf{y}$, our mixture density network (MDN) models the conditional density as

$$p(\mathbf{y} | \mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}) \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_k(\mathbf{x}), \boldsymbol{\Sigma}_k(\mathbf{x})), \quad \sum_{k=1}^K \pi_k(\mathbf{x}) = 1. \quad (1)$$

We define our MDN network as a feedforward neural network with two hidden layers and ReLU activations. The network is trained to produce three things: the mixture weights $\pi_k(\mathbf{x})$, means $\mu_k(\mathbf{x})$, and covariance matrices $\Sigma_k(\mathbf{x})$ for each component.

In more detail, the first head outputs logits, which is then passed through a softmax to produce the mixture weights:

$$\pi_k(\mathbf{x}) = \frac{\exp(a_k(\mathbf{x}))}{\sum_{j=1}^K \exp(a_j(\mathbf{x}))}, \quad \sum_{k=1}^K \pi_k(\mathbf{x}) = 1.$$

The second head outputs the means $\mu_k(\mathbf{x})$ directly, while the third head outputs parameterizes a positive precision matrix

$$\mathbf{\Lambda}_k(\mathbf{x}) = \text{diag}(\text{ReLU}(\mathbf{r}_k(\mathbf{x})) + \epsilon) + \mathbf{L}_k(\mathbf{x})\mathbf{L}_k(\mathbf{x})^\top,$$

where $\epsilon = 10^{-3}$ ensures positive definiteness (so that invertibility is guaranteed). The covariance is $\boldsymbol{\Sigma}_k(\mathbf{x}) = \mathbf{\Lambda}_k(\mathbf{x})^{-1}$.

The network is trained by minimizing the mean negative log likelihood, specifically computing the mixture likelihood with a log-sum-exp expression for numerical stability. For point-prediction RMSE, we use the mixture mean $\sum_{k=1}^K \pi_k(\mathbf{x})\mu_k(\mathbf{x})$. Our hyperparameters include the number of mixture components K (the number of Gaussians in the mixture), the rank of the covariance matrices (1), learning rate (1e-3), batch size (64), and number of epochs (300).

2 Synthetic two-dimensional example

The synthetic data for our 2D example was generated by choosing points from two linear mappings, $y = a_1x$ or $y = a_2x$ with equal probability, and then adding Gaussian noise, $\epsilon \sim \mathcal{N}(0, 0.1)$. This resulted in a multimodal distribution for the output given the input, see Figure 1. The resulting trained model should show a single Gaussian near the intersection of the distributions, and two separate Gaussians that are increasingly far apart as the input moves away from the intersection (if the MDN learns well). Figure 2 does indeed show this behavior. The MDN adequately finds each mode with respect to the data.

Additionally, we tried a range of values for K , see Figure 3, and found that there was no significant improvement for any K greater than 2. This is expected because the data comes from a simple two-modal distribution.

We also compare these K models to a deterministic regressor. This regressor performed similarly to the $K = 1$ MDN. NLL and RMSE are reported for all the aforementioned models in Table 1, when relevant. The best test RMSE is achieved by $K = 2$ at 0.9040, while the best test NLL is achieved by $K = 4$ at -0.2536 .

We observe that training loss (NLL or MSE) over epochs decreases for all models, and that MDN models with $K > 1$ outperform the deterministic and single-Gaussian models. Models with $K \in 2, 3, 4, 5$ perform similarly, with $K = 2$ best on RMSE and $K = 4$ best on NLL.

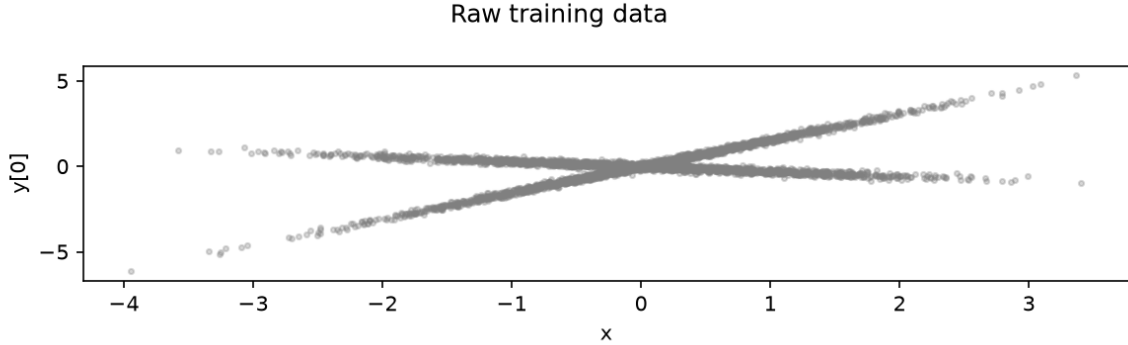


Figure 1: Raw training observations for the synthetic task, showing the multiple output modes associated with the input.

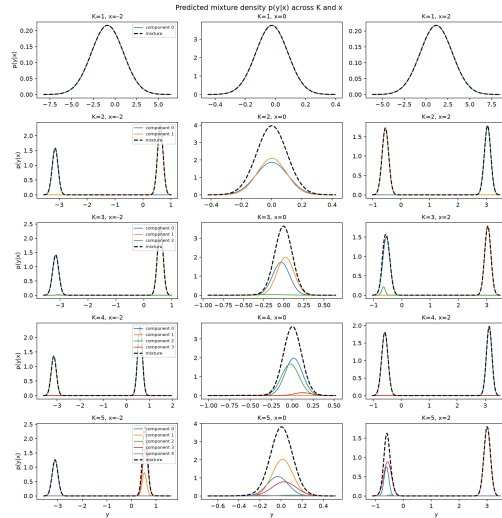


Figure 2: Learned conditional density for the synthetic 2D task over a grid of input and output values. The density should reveal the modes captured by the MDN and how their locations, weights, and widths vary with the input.

3 Higher-dimensional realistic data

We use the Wind Turbine SCADA dataset available on Kaggle as the realistic data set. This data set contains the features described in Table 2. To construct a trainable feature set we decided to drop the Theoretical Power Curve, as we are unsure how the manufacturer generated this feature. We also replaced the raw Date/Time timestamp with cyclical (sine/cosine) encodings of hour-of-day and day-of-year, mapping each periodic time feature onto the unit circle via $\theta = 2\pi t/\text{period}$ (period = 24 for hour, using the fractional hour to account for the dataset’s 10-minute sampling; period = 365 for day-of-year) so that adjacent times (e.g. 24:00 and 00:01) remain close in feature space rather than being separated by the full width of the raw numeric scale. Wind speed and wind direction were kept as direct scalar inputs.

Figure 4 shows each feature plotted against the target, LV ActivePower. From a visual perspective, most features did

Table 1: Synthetic-data performance. NLL is not defined for the deterministic model.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	0.918259	0.915381
Single Gaussian	1	0.833681	0.809800	0.922317	0.906886
MDN	2	-0.213716	-0.222193	0.921555	0.903964
MDN	3	-0.238187	-0.248380	0.920613	0.905017
MDN	4	-0.239954	-0.253606	0.920990	0.907196
MDN	5	-0.239728	-0.235432	0.919976	0.910573

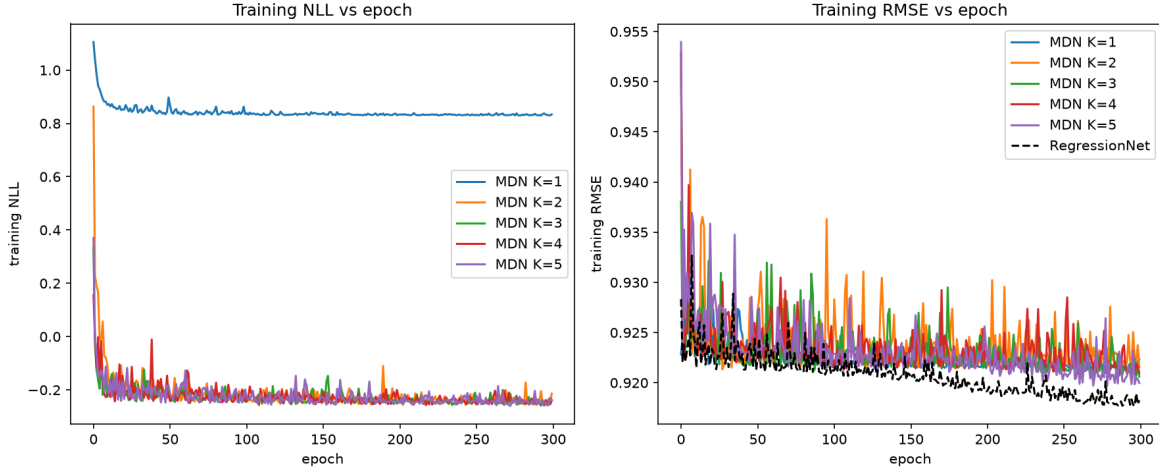


Figure 3: Training objective and RMSE across epochs for the deterministic baseline, single-Gaussian model, and MDNs with different numbers of components.

Table 2: Features of the wind turbine SCADA dataset.

Feature	Unit	Description
Date/Time	–	Timestamp, recorded at 10-minute intervals
LV ActivePower	kW	Power generated by the turbine at that moment
Wind Speed	m/s	Wind speed at the hub height of the turbine
Theoretical Power Curve	kWh	Theoretical power the turbine generates at a given wind speed, as specified by the manufacturer
Wind Direction	°	Wind direction at the hub height of the turbine

not obviously display a multimodal relationship with respect to ActivePower except for Wind Speed. In most cases the entire range of ActivePower can be observed for nearly all feature values, however, Wind Speed does display a multimodal relationship with the target: across the ramp region (roughly 4-13 m/s) and at or near maximum power output (13 m/s and beyond). Wind Direction shows a related but weaker pattern, power appears to switch between near-zero and near-max output across direction bands, in particular straddling 200° , but this on/off behavior is not by itself evidence of a genuinely multimodal conditional distribution. Figure 5 shows the conditional distribution of ActivePower within narrow bands straddling several representative query points: 5 ± 0.5 , 9 ± 0.5 , and 13 ± 0.5 m/s for Wind Speed, $50 \pm 10^\circ$ and $200 \pm 10^\circ$ for Wind Direction. Unlike the scatter in Figure 4, this makes any separation in $P(\text{ActivePower} | x_0)$ explicit as distinct histogram peaks rather than something inferred from the shape of a scatter cloud.

Using the same model architectures, adapted to the higher-dimensional input, and training recipe as the synthetic 2D example (Adam, batch size: 64, epochs: 300), we performed an 80/20 train/test split via a random sampling. We observed the train and test loss (NLL for the MDNs and MSE for the regression model) and RMSE for both model families across K , shown in Table 3 and Figures 6. Training NLL for the MDNs improves fairly consistently as K increases with a small non-monotonic dip at $k = 4$, the same pattern is also observed in the test set. Broadly this training pattern was expected as additional mixture components let the model dedicate separate, narrower gaussian’s to different modes rather than forcing a single gaussian to cover the full spread of outcomes. We believe the small dip at $k = 4$ simply reflects the non-convexity of the optimization rather than a violation of the expectation that increasing k provides more mode coverage. In comparing RMSE, the regression model achieves the best test RMSE of any model (195.93 kW), beating even the best MDN ($K = 5$, 223.37 kW). Since the regression model is trained to directly minimize MSE, this is expected as the MDNs mixture mean is a byproduct of maximizing the likelihood, and MSE is not the objective its training was directly optimizing for. When the MDN correctly discovers separated modes, the mixture mean will fall in the low-density region between the modes which serves as a poor point estimate for either mode. Thus, a model that better captures the real multimodality of the data can therefore look worse under RMSE than one that never attempts to represent different modes.

Comparing the train and test columns of Table 3 further shows that the models generalize differently. The single gaussian model ($K = 1$) has a smaller train test gap (289.54 to 293.17 kW) but by far the worst RMSE overall. This

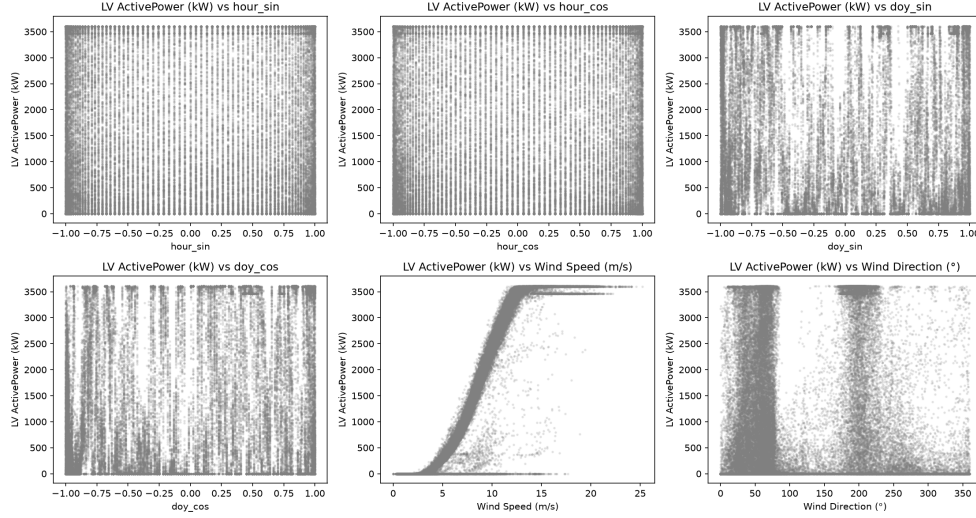


Figure 4: Wind-turbine active power against each available numeric predictor, used to assess multimodality and heteroscedasticity before fitting the models.

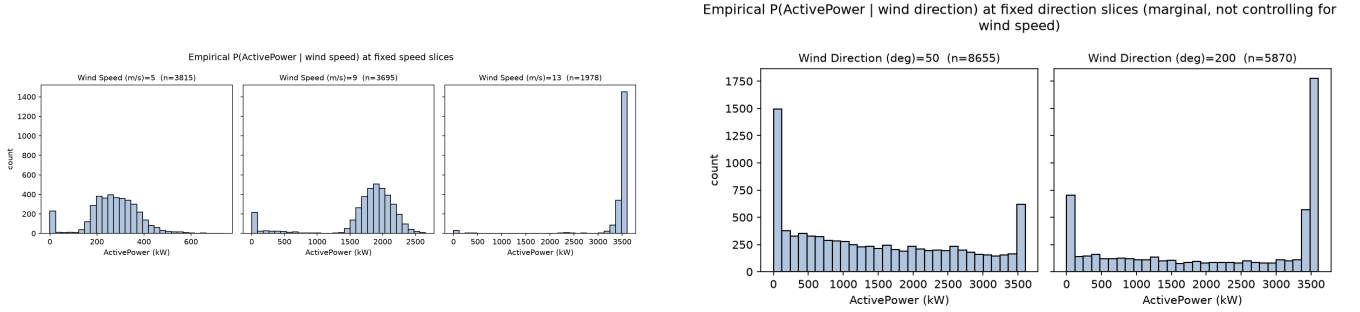


Figure 5: Conditional histograms of ActivePower at fixed slices of wind speed (left) and wind direction (right).

suggests the MDN with $K = 1$ is an underfit model as a single unimodal gaussian cannot represent the multi modality observed in our exploratory data analysis, thus the small increase in test RMSE may appear that it generalizes well, but this is largely because it describes the data poorly to begin with. In contrast, the regression model shows the best overall train RMSE, but the jump in RMSE on the test set suggests it generalizes poorly. The MDNs with $K \geq 2$ sit in between these models, fitting more closely than the single gaussian without the same degree of overfitting observed with the regression model.

Each curve overlay plot, Figures 7 and 8, is produced by a fix-sweep approach: a sequence of point queries in which the feature under inspection is varied across a grid of 200 values spanning its domain, while every other input feature is held fixed at a single representative value (wind speed = 9 m/s, wind direction = 50° , hour-of-day = 12:00, day-of-year = 182) rather than allowed to vary. At each of the 200 points we query the trained MDN and regression models. The resulting means, ± 1 std bands, and regression predictions are plotted as a function of the swept feature. The dashed line is the regression prediction and the colored bands are the ± 1 std bands for each K in the given MDN model.

Table 3: Wind-turbine performance loaded directly from `csvs/wind_turbine_summary.csv`. RMSE is reported in kW; NLL is evaluated on standardized targets and is not defined for the deterministic model.

Model	K	Train NLL	Test NLL	Train RMSE	Test RMSE
Deterministic	—	N/A	N/A	152.436249	195.933411
Single Gaussian	1	-1.931191	-1.296546	289.542419	293.168152
MDN	2	-2.129843	-2.014865	205.125427	231.207932
MDN	3	-2.311052	-2.186236	197.384781	231.579971
MDN	4	-2.293750	-2.171230	210.788620	238.717484
MDN	5	-2.374486	-2.247217	198.114655	223.374252

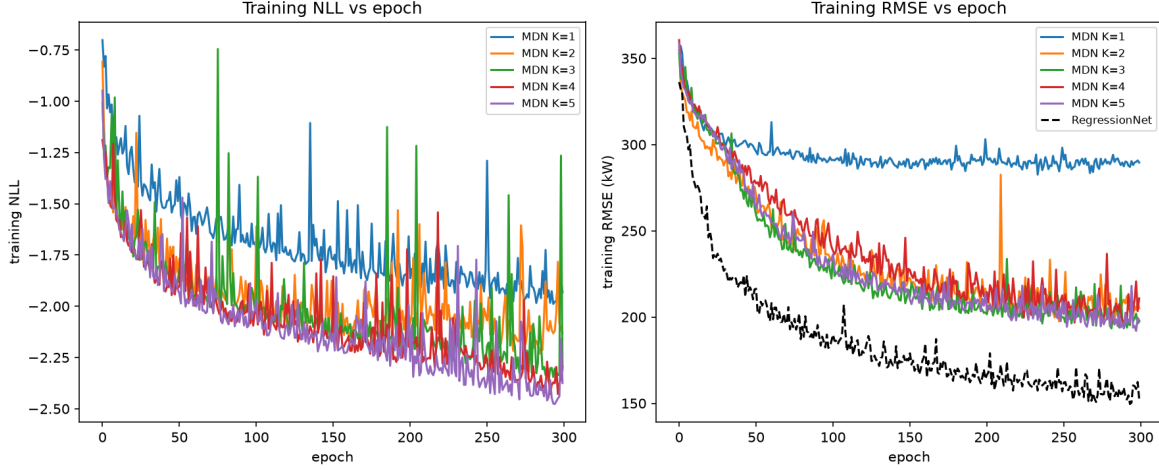


Figure 6: Training and validation/test behavior for the realistic-data models, including the selected checkpoint or stopping point.

Note that each curve overlay plot only reports a single fixed value, for example “Wind Direction ($^{\circ}$) = 50 ± 10 ” in the wind-speed sweep. These values denote the point and tolerance values used to restrict the raw training points that are plotted alongside the model curves, thus each plot only contains a small representative sample of the raw data. Every other feature not named in the figure title is left unfiltered, thus the raw data scatter is not conditioned the same across each curve overlay plot.

The plots let us check feature by feature, whether the MDN’s components are modeling real modes or fitting frivolous gaussians that overall do not contribute strongly to the mixture. Each component k carries a mixture weight $\pi_k(\mathbf{x}) \in [0, 1]$, with $\sum_k \pi_k(\mathbf{x}) = 1$: this is the fraction of predicted probability assigned to that component at input $\mathbf{x}$. A component’s mean can move substantially across a sweep while its weight stays near zero, in which case the gaussian does not contribute strongly to the model’s overall predictive power. So, we read weight and mean together rather than judging multimodality from the banded curves alone.

Wind speed provides the clearest evidence that the MDN discovers different modes, but they are not the modes we hypothesized in the exploratory analysis. Figure 5 motivated wind speed as multimodal because a fixed-speed slice showed evidence of separate clusters, in particular at wind speed straddling 5 and 9 m/s. Checking the same points against the trained models and weights, we do see multiple gaussians with a significant amount of contribution to the overall mixture around these points, however, the dominant components regardless of K seem to fit the main trend line and not the curtailment around 0 nor the two branches near the plateau. For the MDN with $K = 4$ we do see the mean of the orange component roughly fitting the theoretical mode identified in the exploratory analysis, however, at 9 m/s the mean is -16.5 with an overall weight of 0.13% in the mixture. We see a similar weighting to all points in this theoretical mode where $K > 2$.

What the MDNs seem to be fitting, especially for $K = 4$, is an *on-curve* mode that is not obvious through simple visualization. At $K = 4$ and 9 m/s, 59% of the weight sits on the red component and 41% on the green component, both close to the main trend line. $K = 4$ carries the highest average secondary weight of any K across the full speed sweep, and $K = 3/5$ recover a similar on-tend split near the max-power plateau. We read this as a genuine, but unexpected, finding; it seems the previously identified modes did not carry enough weight to significantly impact the training of the mixture. Instead, the primary signal is the concentrated trend line that the regression model also fits.

The other features appeared to show a weaker case of multimodality as well as being strong signals the networks used for learning. Figure 8 displays similar fix-sweep plots for the wind direction and day of year features. Unlike the sweep plots for wind speed both wind direction and day of year MDN components appear to be unstable. In the case of wind speed, components appear to fit similar trends regardless of K , whereas there appears to be more variability across K for wind direction and day of year. These features do seem to have a somewhat stable primary component (orange in $k = 2 - 3$ for both features), however, the secondary components vary wildly. Wind direction’s average secondary weight jumps from 0.04-0.07 at $K = 2, 3$ to 0.27 at $K = 4$ before dropping back to 0.17 at $K = 5$, and day-of-year shows the same $K = 4$ -concentrated pattern (0.06-0.08 at $K = 2, 3$ versus 0.21 at $K = 4$). Taken together, direction and day of year do seem to have a true contribution to training signal, however, there is weak evidence these features contribute to a multimodal relationship with Active Power.

Figure 9 shows the full predicted density $p(\text{ActivePower} \mid \mathbf{x})$ at three representative wind-speed values (5, 9, and 13 m/s) across all five values of K , complementing the curve-overlay sweep by showing the actual shape of the predicted distribution at fixed points rather than only its swept mean and weight. This view further supports the density is largely contained on the wind speed curve and the theoretical mode does not contribute significant weight to the overall mixture. The strongest evidence for multimodality is $K = 4$ at a wind speed of 9 m/s, where the mixture splits nearly evenly (59%/41%) between two components about 99 kW apart. Other points, such as $K = 4$ at 5 m/s (81.5%/15.6%, only 39 kW apart), show a similar but less balanced split, and even at 9 m/s the two components substantially overlap given their spread - the mixture is largely composed of gaussians with high overlap rather than sharply separated modes.

4 Conclusion

On the synthetic task, the MDN beats the regression model outright. Every MDN, including $K = 1$, achieves better test RMSE than the deterministic regression network, with $K = 2$ best on RMSE and $K = 4$ best on NLL. Here representing the two modes directly contributed to overall improved performance.

On the wind-turbine task, the regression network achieves the best test RMSE overall (195.93 kW), beating every MDN including the best of them ($K = 5$, 223.37 kW), while the MDNs still achieve better test NLL than the single-Gaussian model as K increases. Where the MDN represents two components as genuinely separated, the mixture mean falls between them in a region of low density, giving a worse point estimate even though the underlying distribution is better represented than the regression model. So, the MDN’s advantage over regression is consistently distributional (NLL), not a point-accuracy (RMSE). For an MDN to beat a deterministic model on point prediction, the underlying data needs more cleanly separated modes than the wind-turbine data provided here.

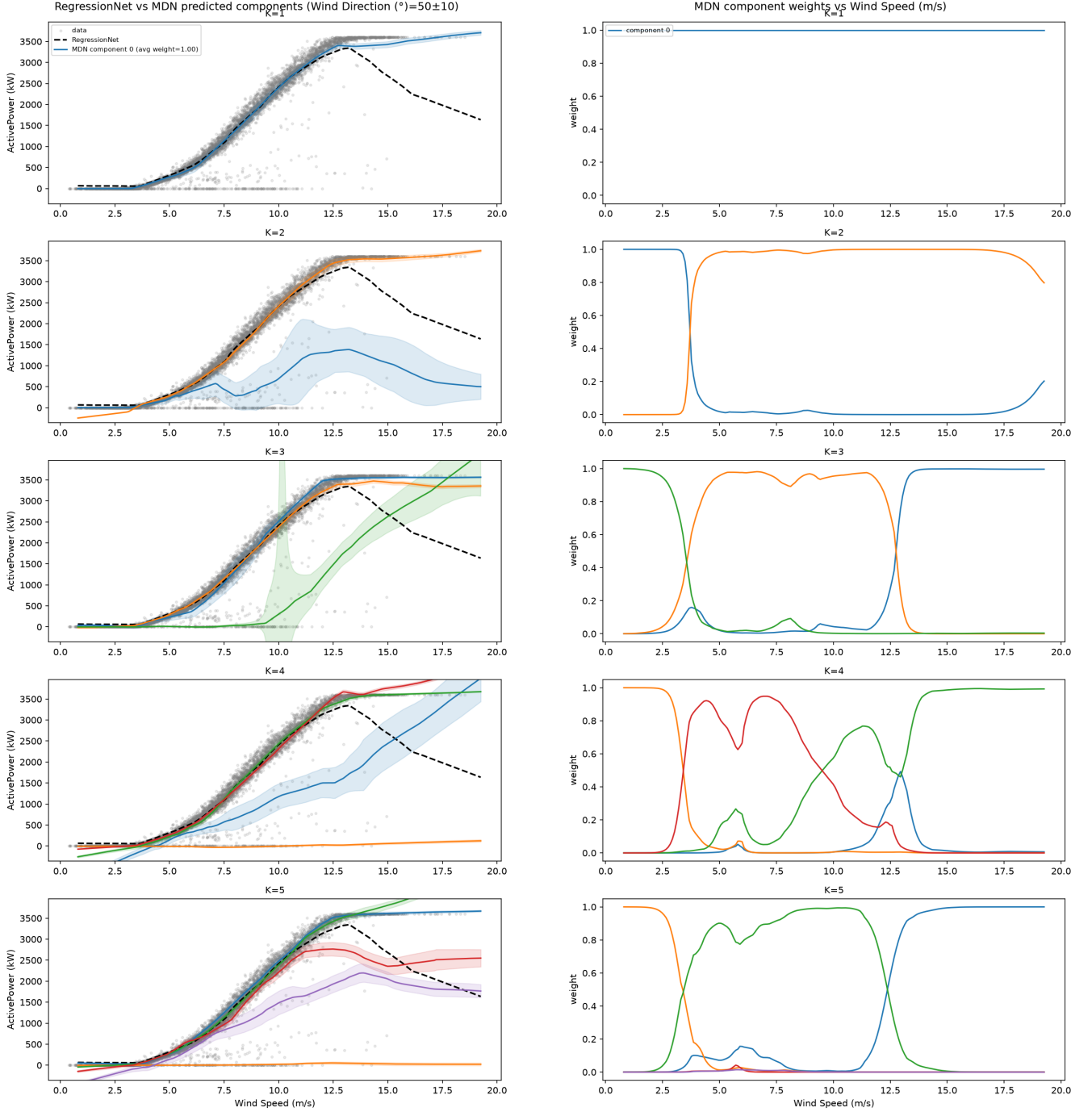


Figure 7: The regression network's predicted curve vs. each MDN's predicted component means (colored, ± 1 std shaded) swept over wind speed (left) with one row per K , overlaid on raw data. Each MDN's predicted component's respective weight values across the sweep (right) shows the weight each component provides across wind speed.

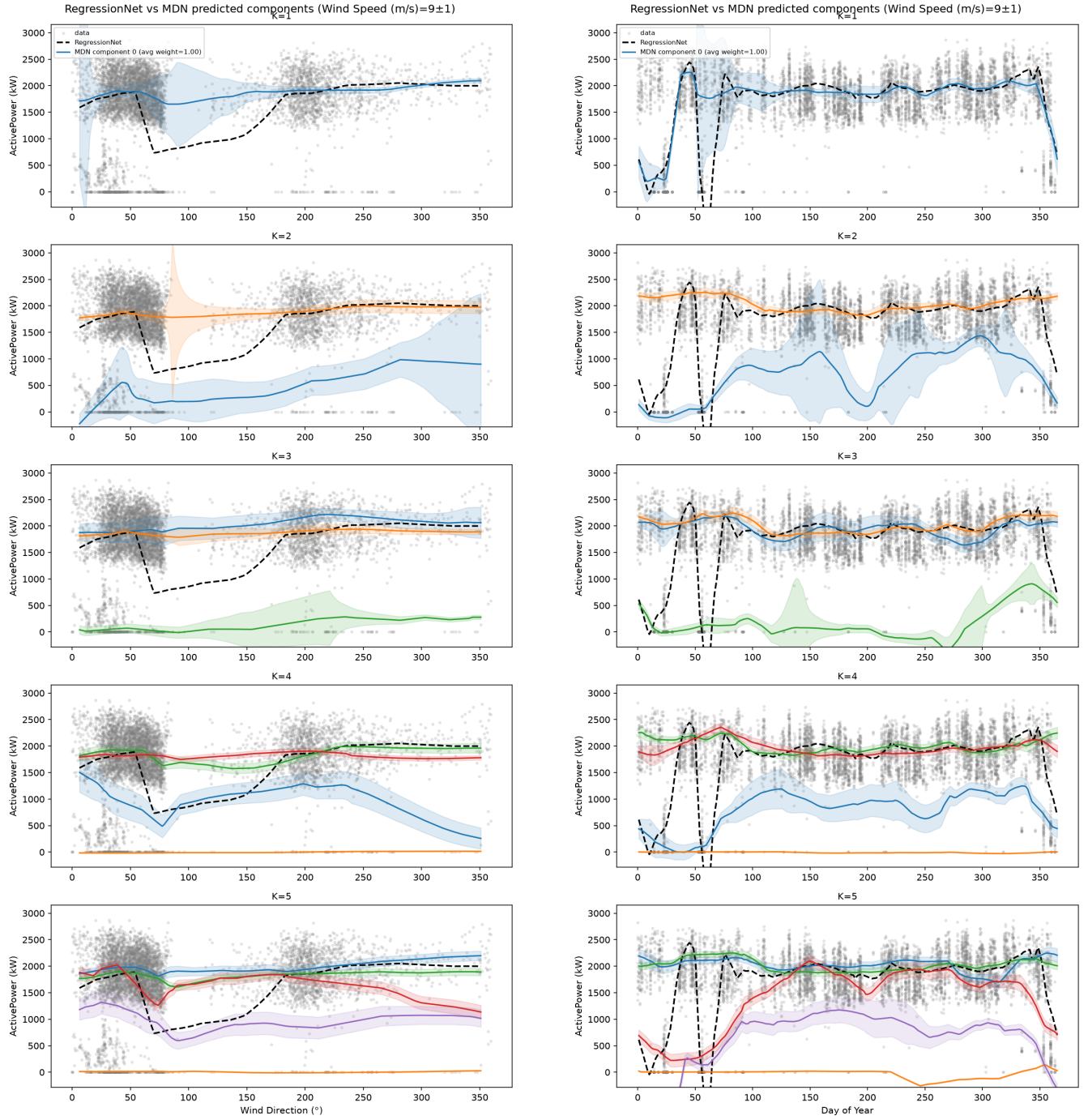


Figure 8: MDN component sweep over wind direction (left) and day-of-year (right).

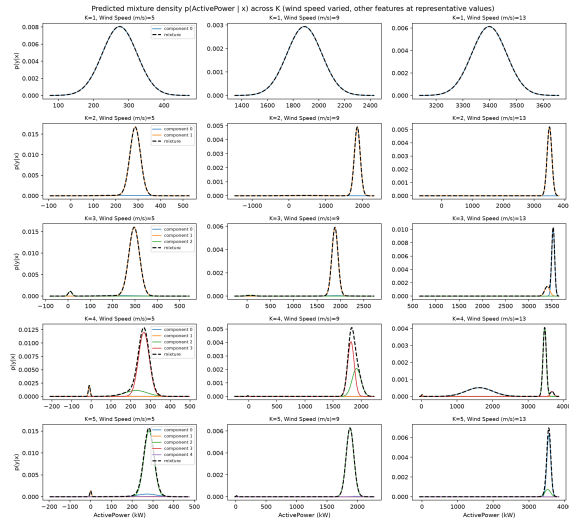


Figure 9: Learned conditional density for the wind-turbine task over a grid of inputs and active-power outputs. Use this view to show heteroscedasticity, multiple operating regimes, and regions where the model assigns diffuse or multimodal probability.